Milestone - 2

Receipt & Invoice Digitizer

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Problem Statement

- Students and businesses struggle to manually extract data from invoices and receipts. Manual entry is time-consuming and error-prone. An automated OCR-based system is needed to convert bill images into structured digital data.
- It extracts key details such as vendor **name, invoice number, date, and total amount**, and stores them securely in a database.

Objective

- Build OCR pipeline using Tesseract
- Extract key fields (date, total, invoice no)
- Convert unstructured text → structured JSON
- Prepare foundation for intelligent document processing

Technology Stack

COMPONENT	TECHNOLOGY
OCR Engine	*Tesseract*
Image Processing	*OpenCV*
Language	*Python*
Environment	*Google Colab*

Flow

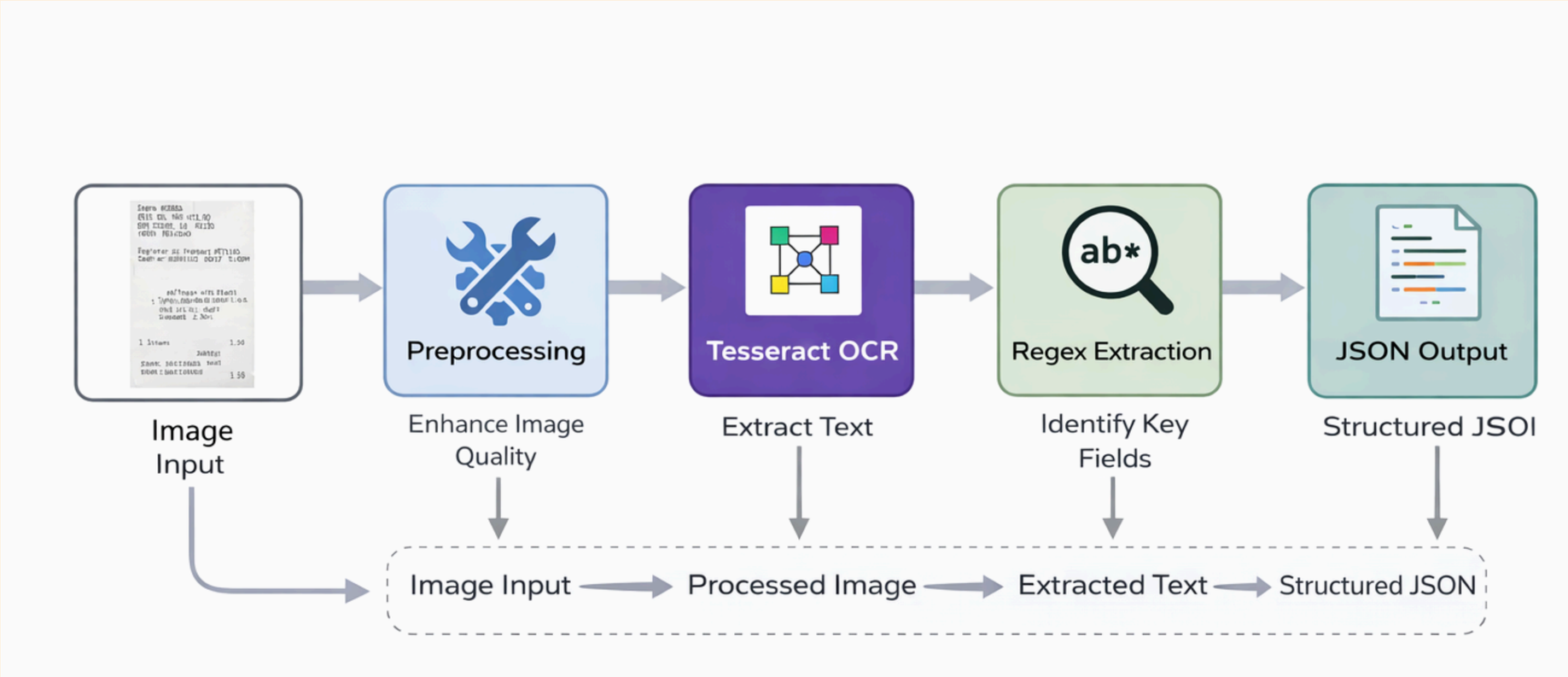
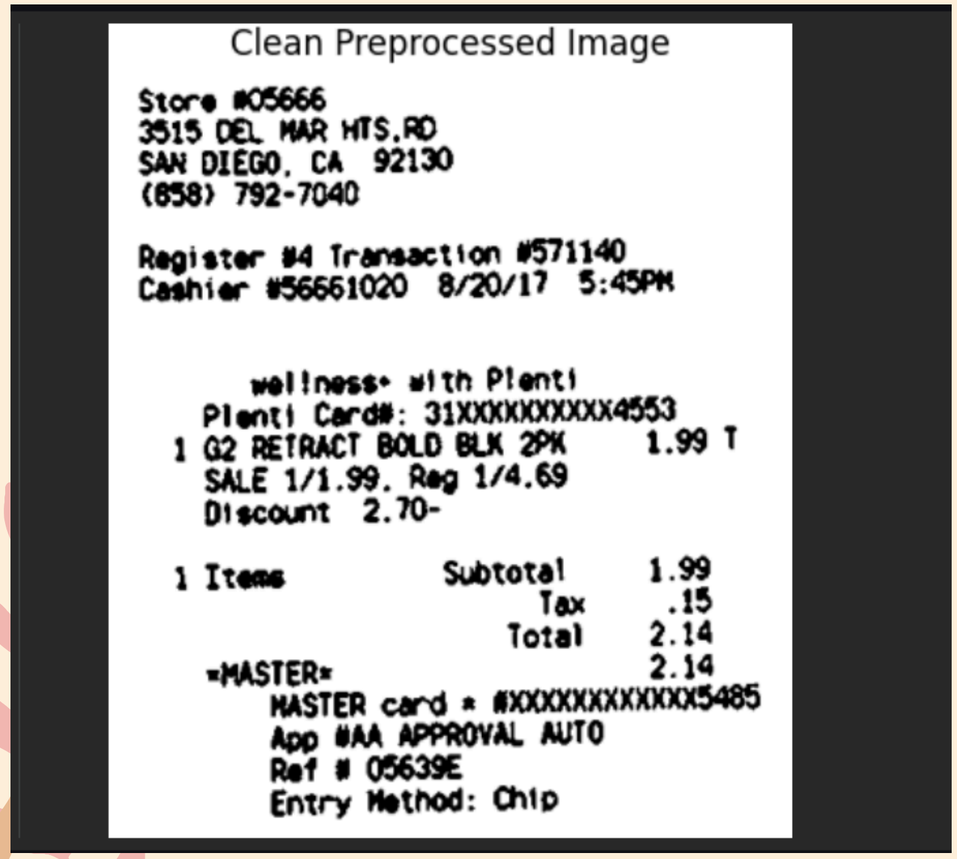


Image Input
Preprocessing
Enhance Image Quality
Tesseract OCR
Extract Text
Regex Extraction
Identify Key Fields
JSON Output
Structured JSOI
Image Input
Processed Image
Extracted Text
Structured JSON

Result




```
••• ===== OCR TEXT =====
    Store MS6E6
    3515 DEL MAR HTS.RO
    SAN DIEGO, CA 92190
    (858) 792-7040
    Register #4 Transaction #571140
    Cashier 856661020 8/20/17 5:45PK
    wel!nesse with Plents
    Plenti Cade: JIXKKOIKNGSSS
    1 G2 RETRACT BOLD BLK PK 1.99 T
    SALE 171.99. Reg 174.69
    Discount 2.70-
    1 Tress Subtotal 1.99
    Tax i]
    Total = 2.14
    =MASTER: 2.14
    WASTER card = MXXKKKKKKKNXKS485
    App WAA APPROVAL AUTO
    Ref # 05639E
    Entry Method: Chip
```

Thank You